



- 8 A program is being developed to implement a game for up to six players.

During the game, each player assembles a team of characters. At the start of the game there are 45 characters available.

Each character has four attributes, as follows:

Attribute	Examples	Comment
Player	0, 1, 3	The player the character is assigned to.
Role	Builder, Teacher, Doctor	The job that the character will perform in the game.
Name	Bill, Lee, Farah, Mo	The name of the character. Several characters may perform the same role, but they will each have a unique name.
Level	14, 23, 76	The skill level of the character. An integer in the range 0 to 100 inclusive.

The programmer has defined a record type to define each character.

The record type definition is shown in pseudocode as follows:

```

TYPE CharacterType
    DECLARE Player : INTEGER
    DECLARE Role : STRING
    DECLARE Name : STRING
    DECLARE Level : INTEGER
ENDTYPE
  
```

The `Player` field indicates the player to which the character is assigned (1 to 6). The field value is 0 if the character is **not** assigned to any player.

The programmer has defined a global array to store the character data as follows:

```
DECLARE Character : ARRAY[1:45] OF CharacterType
```

At the start of the game all record fields are initialised, and all `Player` field values are set to 0

The programmer has defined a program module as follows:

Module	Description
<code>Assign()</code>	- called with two parameters: - an integer representing a player - a string representing a character role - search the <code>Character</code> array for an unassigned character with the required role - If found, assign the character to the given player and output a confirmation message, for example: "Bill the Builder has been assigned to player 3" - If no unassigned character with the required role is found, output a suitable message.

